CURRICULUM COMPANION : TECHNICAL LEARNING MODULES & FREE COURSES

Week 4 Learning Materials : Productionization & Capstone Microservices

FastAPI Lifespans, Hardened Docker, Prometheus Metrics, CI/CD & 100% Free Courses

TARGET AUDIENCE	ESTIMATED READING TIME	COURSE COST	PRIMARY OBJECTIVE
ML Interns (Capstone Phase)	45-55 Minutes	100% Free (Zero Paywalls)	Production Microservice Rigor

01 Pre-Warmed Lifespan Architecture with FastAPI

MODULE 1

In production, machine learning microservices must never reload model weights from disk or re-initialize neural sessions per request. Use FastAPI modern asynchronous lifespan context managers to load models into memory during boot and warm up computation graphs:

```
# Lifespan context manager for zero cold-start production inference
from contextlib import asynccontextmanager
from fastapi import FastAPI
@asynccontextmanager
async def lifespan(app: FastAPI):
    app.state.model = onnxruntime.InferenceSession("model.onnx")
    app.state.model.run(None, dummy_warmup_tensor) # Warm up JIT graphs
    yield
    app.state.model = None # Graceful teardown
app = FastAPI(lifespan=lifespan)
```

02 Defensive Edge-Case Verification & Negative Testing

MODULE 2

Toy projects test only the happy path. Enterprise software is defined by how gracefully it handles adversarial, malformed, or boundary inputs without throwing unhandled HTTP 500 server crashes:

ATTENOMY DEFENSIVE ENGINEERING STANDARD

1. Negative Test Ratio (>60%): More than 60% of your automated Pytest assertions must test failure states: empty payloads, oversized requests, corrupt base64 strings, and disconnected databases.

2. Pydantic Strict Validation: Use Pydantic v2 `Field(..., ge=0)` and custom field validators to catch schema violations at the gateway layer, returning clean HTTP 422 JSON errors.

3. Graceful Degradation (Circuit Breaking): If external dependencies (e.g. Redis cache) become unreachable, catch timeouts and fail open to baseline computation rather than crashing.

03 Hardened Multi-Stage Containerization (Non-Root User) MODULE 3

Running production Docker containers as the default `root` user violates security standards. Multi-stage builds compile dependencies in a build stage, copy only wheels into a slim runtime, and drop privileges to a dedicated non-root user:

```
# Hardened multi-stage Dockerfile snippet
FROM python:3.10-slim AS builder
WORKDIR /build && pip install --no-cache-dir --user -r requirements.txt
FROM python:3.10-slim AS runtime
RUN groupadd -g 10001 attenomy && useradd -u 10001 -g attenomy -s /bin/bash -m attenomy
COPY --from=builder /root/.local /home/attenomy/.local
USER attenomy
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

04 Observability, Tail Latencies & Prometheus Metrics MODULE 4

Mean latency hides critical tail spikes. Production monitoring requires histogram quantile tracking to measure median (p50), 95th percentile (p95), and 99th percentile (p99) response latencies:

CORE PROMETHEUS METRICS SCHEMA

1. Counter (`cortex_requests_total`): Cumulative request count partitioned by HTTP method, route, and status code.
2. Histogram (`cortex_latency_seconds`): Exponential duration buckets (0.005s, 0.010s, 0.025s, 0.050s, 0.100s) to compute accurate p95 tail latency.
3. Gauge (`cortex_active_requests`): Real-time in-flight request gauge indicating server concurrency load.

05 Automated GitHub Actions CI/CD Regression Gates MODULE 5

Continuous Integration enforces automated code quality before human review. Every pull request targeting `staging` or `main` must pass automated Ruff linting, security scans, and test suites with zero warnings:

```
# GitHub Actions CI workflow triggers on pull requests
name: Cortex Microservice CI
on: [pull_request]
jobs:
  verify:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - run: pip install -r requirements.txt && ruff check . && pytest -v --cov
```

06 Curated 100% Free Production Engineering Courses & Labs

ZERO PAYWALLS

All recommended courses and technical resources below are completely free, open-access, and require zero credit cards or paid subscriptions:

RESOURCE / COURSE	AUTHOR / PLATFORM	KEY TOPICS COVERED	ACCESS FORMAT
FastAPI Official Documentation	Sebastián Ramírez	Asynchronous routes, lifespan managers, dependency injection, Pydantic v2	Free Official Docs
Docker for Beginners & Non-Root	FreeCodeCamp & Docker Docs	Multi-stage builds, non-root users, volume binds, compose orchestration	Free 4-Hour YouTube Course
Prometheus Monitoring Mastery	Prometheus.io	Metric types (Counters, Histograms), PromQL quantile queries, alerting	Free Interactive Guides
GitHub Actions CI/CD Lab	GitHub Skills	Automated workflows, secret management, matrix builds, artifact caching	Free Hands-On Labs
Google Engineering Practices	Google Open Source	Code defense standards, Architecture Decision Records (ADRs), code health	Open Public Web Guide

07 Week 4 Reading & Capstone Delivery Checklist

VERIFICATION CHECKLIST

- ☐ **FastAPI Lifespan Docs:** Implement lifespan context manager with pre-warmed weights.
- ☐ **Locust / cProfile:** Profile concurrent requests to prove p95 latency < 50ms on CPU.
- ☐ **Evidence-Driven PR:** Fill out all 5 EDPR sections with test logs and Loom demo link.
- ☐ **Multi-Stage Dockerfile:** Enforce unprivileged user attenomy (UID 10001) in runtime image.
- ☐ **Prometheus /metrics:** Instrument FastAPI with request counters and duration histograms.
- ☐ **2-Minute Defense Prep:** Prepare to defend transaction boundaries and error recovery live.

08 Attenomy Engineering Standard : The 2-Minute Code Defense

FINAL GATE

Autonomous Comprehension: Developers are not evaluated on typing syntax; developers are evaluated on system comprehension, defensive guarantees, and operational clarity. Every contributor must explain every line of committed code during the live 2-minute walkthrough with executive leadership.
Zero Fluff: If you cannot explain the memory allocation, concurrency safety, or failure mode of an AI-generated block, the PR will be rejected. Verify thoroughly, profile rigorously, and ship production software.